ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** floods	Too much water
2	**(c)** clear sunny weather	Sinking air
3	**(c)** Sweat evaporates slowly into air already full of water vapour	High vapour
4	**(a)** Nakshatras - 27 divisions of the Moon's path	Traditional knowledge
5	**(d)** carbon footprint	Less fuel
6	**(d)** the land-sea temperature difference is smaller at night	Difference small
7	**(c)** Cycling to school	Low emission
8	**(c)** monsoon forecasts help farmers plan	Planning
9	**(d)** 4	0.04% of 10,000
10	**(c)** carbon dioxide	Greenhouse gas

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorbs harmful UV (1); Protects living things (1).	1+1
12	Gravity is weak (1); Air is very thin (1).	1+1
13	Livestock lost in water (1); Sheds and fodder destroyed (1).	1+1
14	Slow heating and cooling (1); Moderates temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Land heats faster than sea (1); Low pressure over land, high over sea (1); Moist winds blow in (1).	1+1+1
16	Six seasons of two months (1); Based on the division of the year (1); Vasanta to Shishira (1).	1+1+1
17	Gases trap heat (1); Human activity raises levels (1); Planet warms (1).	1+1+1
	OR: Rising temperature (1); Melting glaciers and sea rise (1); More floods and droughts (1).	1+1+1
18	Large branches sway (1); Umbrellas hard to use (1); Higher than light breeze (1).	1+1+1
	OR: Contains water vapour and clouds (1); Rain, fog and hail occur (1); Air we breathe is here (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Insolation affects temperature (1); Humidity and precipitation (1); Pressure gives wind (1); Wind speed categories (1); Affects climate (1).	1+1+1+1+1
	OR: Save energy and water (1); Public transport (1); Renewable energy (1); Plant trees (1); Policies like NMM and Mission Mausam (1).	1+1+1+1+1
20	Troposphere for life and weather (1); Stratosphere for ozone (1); Mesosphere for meteors (1); Thermosphere for radio and aurora (1); Exosphere for satellites (1).	1+1+1+1+1
	OR: Height: 12, 50, 80 km (1); Temperature falls, rises, falls (1); Troposphere weather, stratosphere ozone, mesosphere meteors (1); Boundaries (1); Role for aviation and life (1).	1+1+1+1+1

Mistakes to avoid:

- **Weather** is the day-to-day state of the atmosphere; **climate** is the average weather over 30 years or more. A single rainy day is not climate.
- Sea breeze blows from sea to land in the **day** (land is warmer, low pressure on land); land breeze blows from land to sea at **night** and is weaker. Do not reverse them.
- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.

- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.
- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Moist sea winds blow towards the heated low-pressure land	Moisture laden
2	**(a)** troposphere	Weather layer
3	**(a)** a north-easterly wind	Named from origin
4	**(c)** little weather turbulence	Smooth flying
5	**(c)** 99%	78 + 21
6	**(c)** 27 parts	Moon's path
7	**(b)** The Moon's phase	Not directly
8	**(c)** cloudy and rainy weather	Rising air
9	**(c)** drought	Low rainfall
10	**(d)** stratosphere	Ozone absorbs UV

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	SW winds come from sea (1); NE winds come from land (1).	1+1
12	Insolation decreases toward poles (1); Torrid, temperate and frigid zones (1).	1+1
13	Absorb CO2 (1); Reduce soil erosion (1).	1+1
14	Ozone absorbs UV radiation (1); Heats the layer (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repair embankments (1); Desilt rivers (1); Early warnings and drills (1).	1+1+1
16	NE monsoon crosses Bay of Bengal (1); Picks moisture (1); Rain in Tamil Nadu and Andhra (1).	1+1+1
17	Weather decides daily tasks (1); Climate decides crops (1); Monsoon governs sowing (1).	1+1+1
	OR: Low: warm rising air, cloudy and wet (1); High: cold sinking air, clear and sunny (1); Wind moves high to low (1).	1+1+1
18	Krishiparashara and Brihatsamhita observed sky (1); Modern models like NMM (1); Both aid farmers (1).	1+1+1
	OR: Old embankments (1); Houses near river (1); Silt and late alerts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Land heats faster in summer (1); Low pressure inland, high over sea (1); Moist winds blow (1); Winter: reverse (1); Rain pattern follows (1).	1+1+1+1+1
	OR: Six ritus (1); Krishiparashara and Brihatsamhita (1); Meghadutam (1); Arthashastra rainfall records (1); Link with farming (1).	1+1+1+1+1
20	Fossil fuels (1); Deforestation and pollution (1); Floods, droughts (1); Melting glaciers (1); Impact on health and farming (1).	1+1+1+1+1
	OR: Agriculture depends on rain (1); Excess causes floods (1); Weak causes drought (1); Water and economy (1); Forecasting helps (1).	1+1+1+1+1

Mistakes to avoid:

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- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.
- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Cooler air from the sea moves in to replace rising warm air over land	Pressure difference
2	**(b)** altitude changes the local climate	Local climates
3	**(a)** mentions the onset of the monsoon over central India	Cloud messenger
4	**(b)** reversal of winds	Wind reversal
5	**(b)** fewer trees remain to absorb it	Carbon sinks lost
6	**(d)** air pressure and density fall with height	Pressure drops
7	**(b)** precipitation	Falling ice
8	**(b)** climate	Long-term
9	**(d)** Winds pick up moisture over the Bay of Bengal before reaching it	Crossing the sea
10	**(c)** Melting glaciers	Water added

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Maximum air column above (1); Pressure falls with height (1).	1+1
12	Based on long observation (1); Used by farmers (1).	1+1
13	Increases river capacity (1); Reduces flood risk (1).	1+1
14	No clouds or weather (1); Calm, smooth flying (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	High humidity brings uncomfortable weather (1); Low pressure brings clouds (1); High pressure brings clarity (1).	1+1+1
16	Falls in troposphere (1); Rises in stratosphere (1); Falls in mesosphere, then rises in thermosphere (1).	1+1+1
17	Stratosphere calm and cloud-free (1); Troposphere has weather and turbulence (1); Safer, smoother flight (1).	1+1+1
	OR: Save electricity (1); Use public transport (1); Avoid plastic (1).	1+1+1
18	Solar wind particles reach the atmosphere (1); Guided by magnetic field to poles (1); Gases glow (1).	1+1+1
	OR: Crops and livestock lost (1); Roads and bridges damaged (1); Waterborne diseases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rain and western disturbance (1); Swollen rivers (1); Weak embankments (1); Buildings near rivers (1); Silt and late warnings (1).	1+1+1+1+1
	OR: IMD and NMM (1); Models and observation (1); Mission Mausam (1); Benefit to farmers (1); Traditional methods also exist (1).	1+1+1+1+1
20	Day sea breeze (1); Night land breeze (1); Pressure difference (1); Moderate climate (1); Fishermen use them (1).	1+1+1+1+1
	OR: Meaning (1); Time scale (1); Elements: temperature, humidity, pressure, wind, precipitation (1); Examples from India (1); Importance (1).	1+1+1+1+1

Mistakes to avoid:

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